CPE 334 SOFTWARE ENGINEERING

Porterest

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Overview

Porterest is a web-based platform designed for students to upload, organize, and showcase their academic portfolios in a visually appealing Pinterest/Medium-style interface. The platform allows viewers to browse, search, and explore student work easily.

Problem

- Students lack a centralized and aesthetically consistent platform to showcase academic work.
- Traditional portfolio formats are difficult to manage and share.
- Teachers, recruiters, and peers need an intuitive way to view student work.

Overview

How the System Works

- Users register through Email or Google Sign-In.
- Students upload portfolios with titles, descriptions, and images.
- Portfolios are displayed in a responsive thumbnail grid.
- Users can edit or delete their own portfolios.
- Admins manage user roles and account status.

Main Objectives

- Provide a simple, fast, and visually engaging portfolio platform.
- Support secure user authentication and role management.
- Ensure high-performance browsing, searching, and rendering.
- Enable efficient portfolio discovery and public viewing.

Project Timeline

Sprint	Week	Goal	Tasks	Deliverables
1	1	Project Setup & Design	- จัดทำ SRS Document - เก็บ Requirement - ทำ Wireframe/Prototype	Draft SRS + Wireframe
1	2	Core Auth	- Authentication (Register/Login) - Database Setup - Basic UI Layout	Prototype + Auth Module
2	3	Portfolio (Part 1)	- Portfolio Create/Upload - Portfolio View	Portfolio Create/View
2	4	Portfolio (Part 2) + QA	- Portfolio Edit/Delete - Portfolio Browsing - Unit Test (Portfolio)	Full Portfolio Module + Test Report
3	5	Admin (Part 1)	- Admin Dashboard UI - Role Management	Admin Dashboard + Roles
3	6	Admin (Part 2) + Security	- User Management - Access Control - Unit Test (Admin)	Admin Features + Test Report
4	7	Extra Functions (Part 1)	- Search Function - Email Notification	Search + Email Features
4	8	Extra Functions (Part 2) +intergration	- File Storage - Integration Test - Deploy to Staging	Extended Features + Staging Deploy
5	9	Validation & System Test	- Validation - Regression Test - Fix Major Bugs	Validation Report
5	10	Final Delivery	- User Feedback & Fix - Final QA - Deployment (Production)	Final Web App + Documentation

Project Details

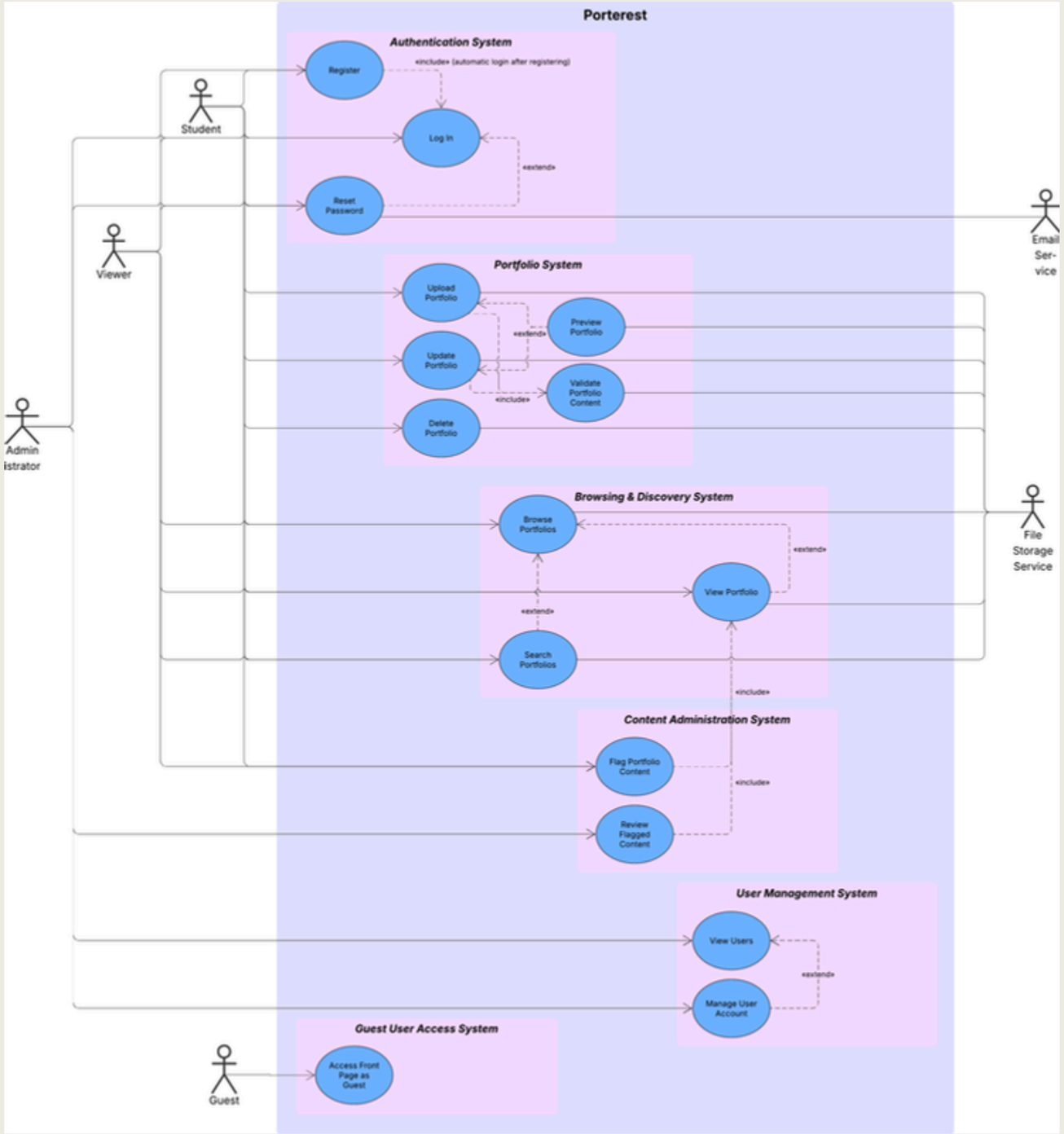
Features

- Authentication & Account Access : User registration, User login, Password reset, Logout, Auto-redirect based on role after login and Logout for all user types
- Role-Based User Homepages : Student Home, Viewer Home and Admin Dashboard
- Portfolio Management (Student Only) : Upload / Edit / Delete
- Portfolio Browsing and Search : Browse portfolio grid, Search portfolio, View portfolio detail
- User management (Admin Only) : Role Management, Account Status Management, User Management
- User Profile : View Profile, Edit name, Edit profile picture

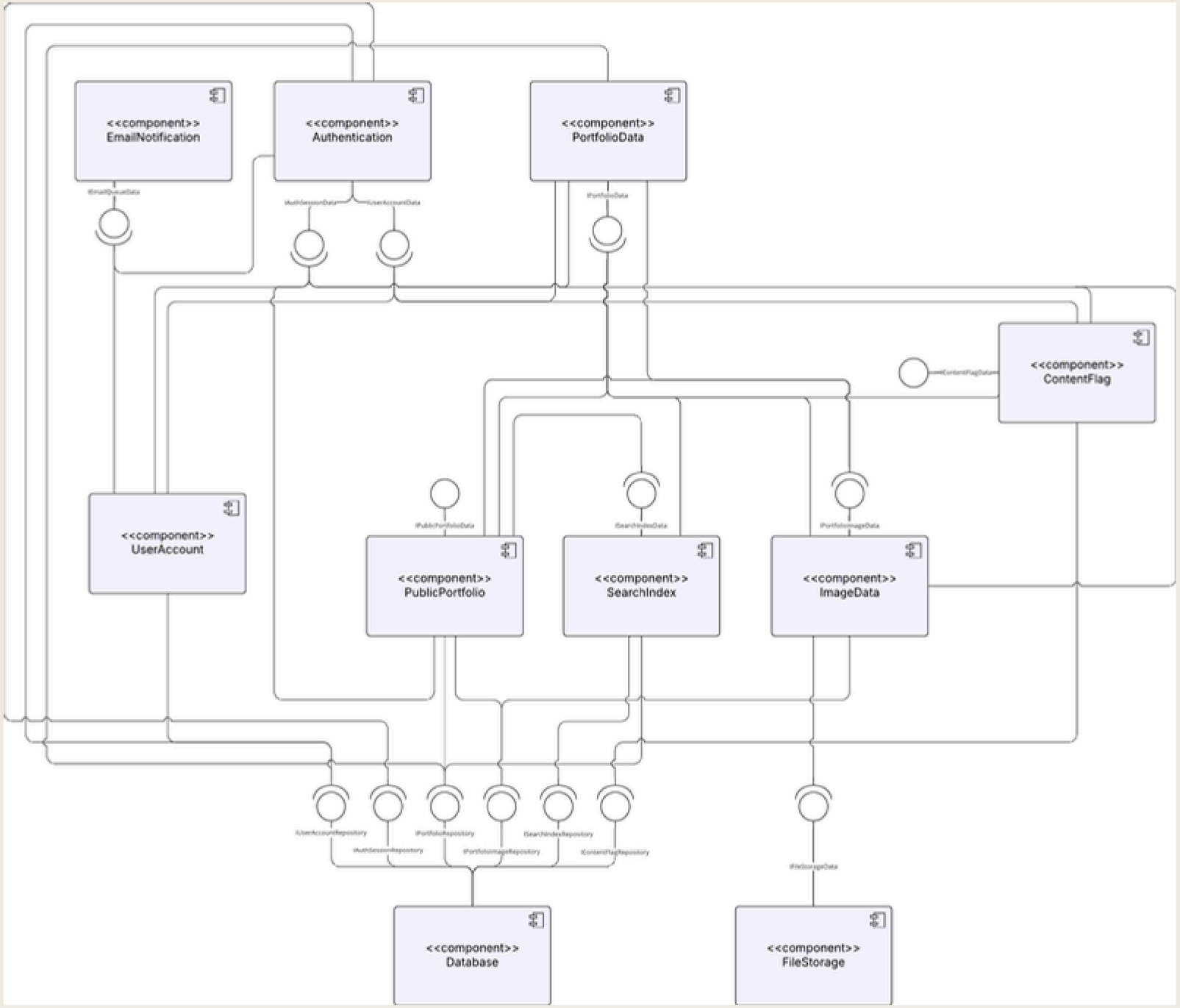
MVP (Minimum Viable Product)

- Authentication : Registration, Login, Logout
- Portfolio Management : Upload portfolio, Edit portfolio, Delete portfolio, Browse portfolio
- Basic Portfolio Browsing : Portfolio Thumbnails Display in grid layout, Clicking a portfolio thumbnail opens the Portfolio Details page

Use Case Diagram



Component Diagram



Role

Features Product owner : Thanawat Prahatchai 66070505209

Scrum master : Ratchanon Tiptamool 66070501045

Developers 3 team

Team A : Kanokkate Laotawee 66070505201
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Team C : Titi Tiyachaipanich 66070501026
Thammapon Chaengpatee 66070501076

Architecture Design

Frontend (Vite + React)

- Renders all UI pages (Login, Portfolio, Admin Dashboard)
- Communicates with backend through REST API
- Handles client-side routing and state management
- Uploads images directly to Supabase Storage

Backend (Elysia.js on Bun)

- Acts as the main API gateway
- Implements business logic:
  - Authentication (if not using Supabase Auth)
  - Portfolio management
  - User management (Admin)
  - Role-based authorization
- Validates requests before interacting with Supabase

Architecture Design

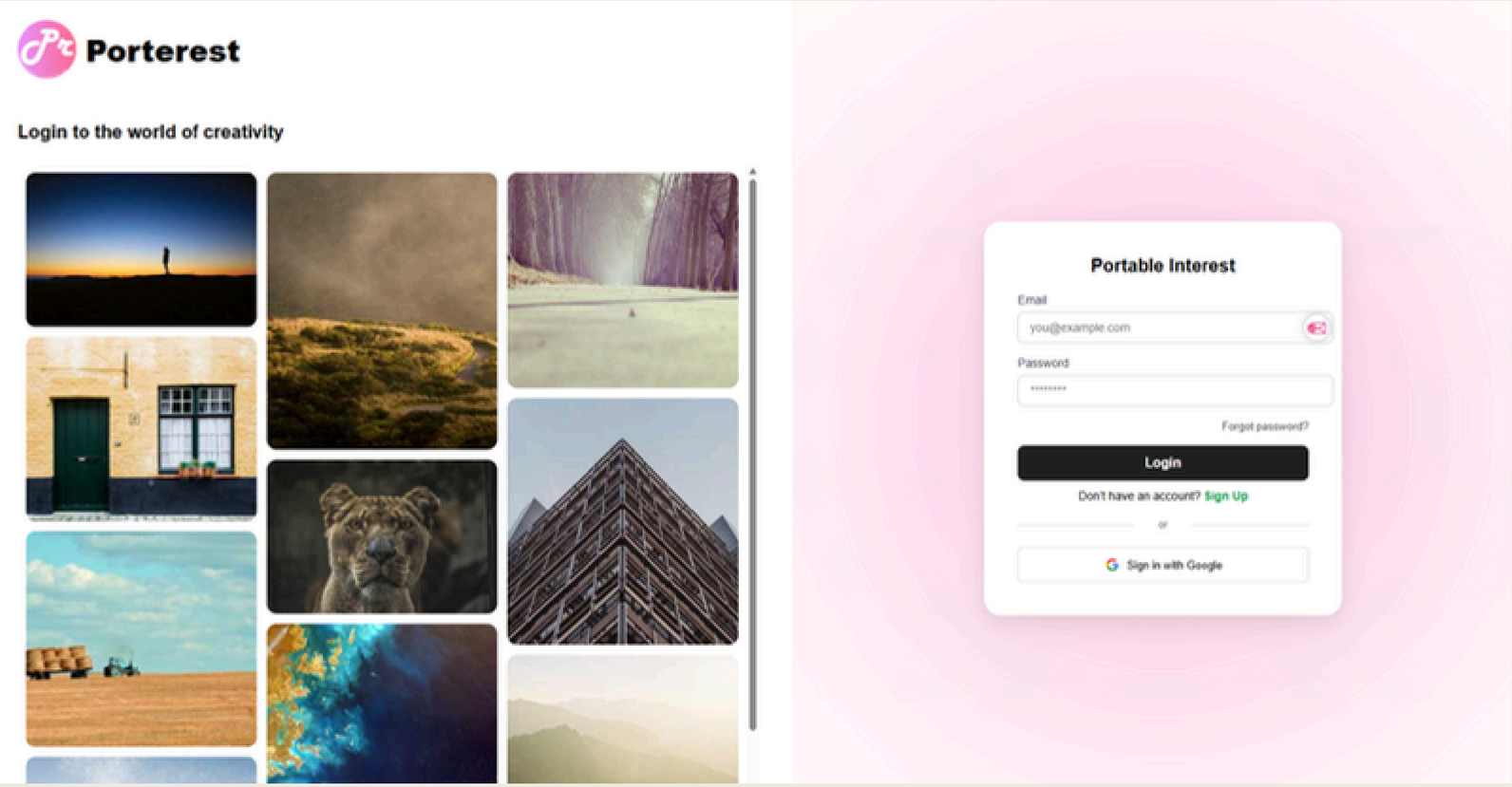
Supabase (Database + Auth + Storage)

- PostgreSQL stores:
  - Users
  - Portfolios
  - Roles
  - Activity logs
- Supabase Storage stores:
  - Portfolio images
- Supabase Auth handles:
  - Email/password sign-in
  - Google OAuth sign-in
  - Token verification

External Services

- Google OAuth for authentication

Demo



video demo

Problem&Solution

The Problem we found.

- Running in the local environment
- Unfamiliar programming language
- The task does not finish within the deadline.

How we solved it.

- After thoroughly checking and researching, we create the document to help with installation.
- After a lot of trial and error, each member shared a lot of info and techniques in the meeting.
- We decide based on the task's priority. Whether to move it to the next sprint or reduce the scope of the task.

Conclusion

We successfully created our Porterest project, which includes all of our essential functions such as Authentication, User Homepages, Portfolio management, and Admin Privilege. Our UX workflow is also validated. The outcome of this project is a robust, secure, and highly functional platform that successfully met all core objectives.

lesson learned

- The critical importance of prioritizing documentation in each step, such as SETUP document.
- Be flexible in every situation; the most important thing is to always prepare for the worst-case scenario.
- Solid structure is a key to a stable, and smooth progression.

Conclusion

future development

- Focused on high-value enhancements, including the integration of analytics dashboards for students to track portfolio impact.
- The expansion of interactive media support for dynamic project displays.
- Create the connection module to help students find more opportunities. And link it with the university's criteria, or the company's criteria.
- Add more variety of feature relate to education.